



FloodSight

AI

Disaster Intelligence & Response Platform

Real-time flood disaster analysis using satellite imagery, multi-agent AI pipelines,
GPU-accelerated object detection and MCP orchestration.

TEAM	MEMBER	ROLE	CHALLENGE
Team-1097	Jeyaprakash	Lead Dev + AI	Disaster Mgmt

Problem
& Context

1 in 5
people globally
live in flood
risk zones

72hrs
critical window
for rescue
operations

80%
of disasters
are flood
related in India

Problem Statement

During floods, emergency responders lack real-time, AI-powered tools to assess flood severity, locate stranded people, estimate population impact, and generate rescue plans — all from a single satellite image in seconds.

Current tools: manual GIS analysis taking hours. No automated AI pipeline exists for India.

Target Users / Stakeholders

- **NDRF Teams** — Real-time rescue planning from satellite feeds
- **State Disaster Mgmt** — District-level flood impact assessment
- **ISRO / IMD** — Automated satellite image intelligence
- **Humanitarian NGOs** — Rapid resource allocation and relief

Why This Problem Matters

- India loses \$3–14B annually to flood damage (World Bank, 2023)
- Bihar / Assam floods displace 2–8 million people every monsoon season
- First 72 hours are critical — current manual assessment takes 4–12 hours
- AI-powered triage can reduce response time by 80% and save thousands of lives

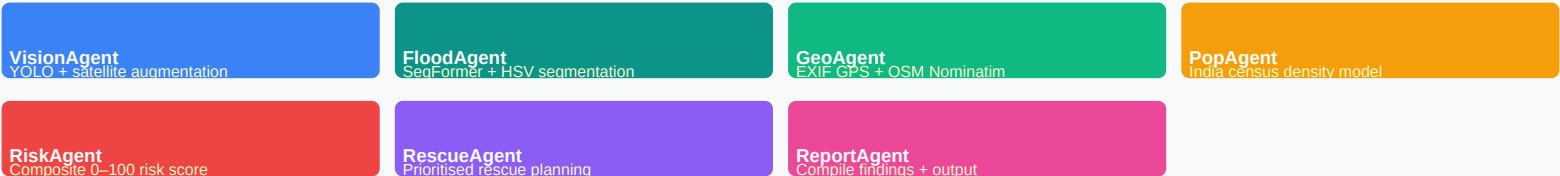
Mapped Challenge: AI for Disaster Management & Emergency Response



Solution Architecture



7-Agent MCP Pipeline



Technology Stack:



GPU	AMD Instinct MI300X
VRAM	8 GB HBM3
Backend	ROCm 7.2
Framework	PyTorch 2.x
Runtime	CUDA-compat

Models, Datasets & Performance

Models Used

Model	Task	Weights	Inference Mode
YOLOv11n	Object Detection (80 COCO classes)	yolo11n.pt (pre-trained)	AMD ROCm GPU
SegFormer mit-b2	Semantic Segmentation	nvidia/mit-b2 (HuggingFace)	AMD ROCm GPU
Qwen3-8B-Instruct	LLM Narrative Reports	vLLM hosted locally	AMD ROCm GPU
OpenCV HSV	Satellite Water Mapping	Rule-based (no training)	CPU fallback

Datasets & Training

- No fine-tuning performed — all models used as pre-trained (off-the-shelf).
- COCO dataset knowledge embedded in YOLOv11n (80 classes, 330K images).
- ADE20K knowledge in SegFormer mit-b2 (150 classes, 20K images).
- Satellite augmentation uses rule-based OpenCV (no training data required).

Performance Metrics



Impact &
Demo

80%
faster
response
time

< 5s
full
analysis
per image

7
AI agents
in one
pipeline

Impact & Demo Summary

Expected Impact



- From 4–12 hrs manual assessment to under 5 seconds automated analysis
- One analyst can process 100s of images in minutes using batch mode
- Deployable for any flood-prone region with satellite/drone imagery
- Early accurate risk scores trigger faster NDRF deployment
- Responders get prioritised rescue plans with resource lists instantly

Key Differentiators

Complete output: flood map + risk score + rescue plan + population + report in one call

Demo Flow — What the Jury Should Notice

